

**1. What do you mean by Flood Frequency Analysis (FFA)? Explain the method of FFA and the importance of FFA in hydropower development.**

Answer:

Flood Frequency Analysis (FFA) is a method used to estimate the magnitude of flood discharge corresponding to a specified return period, such as 10, 50, 100 or 1000 years, using a series of historical flood records.

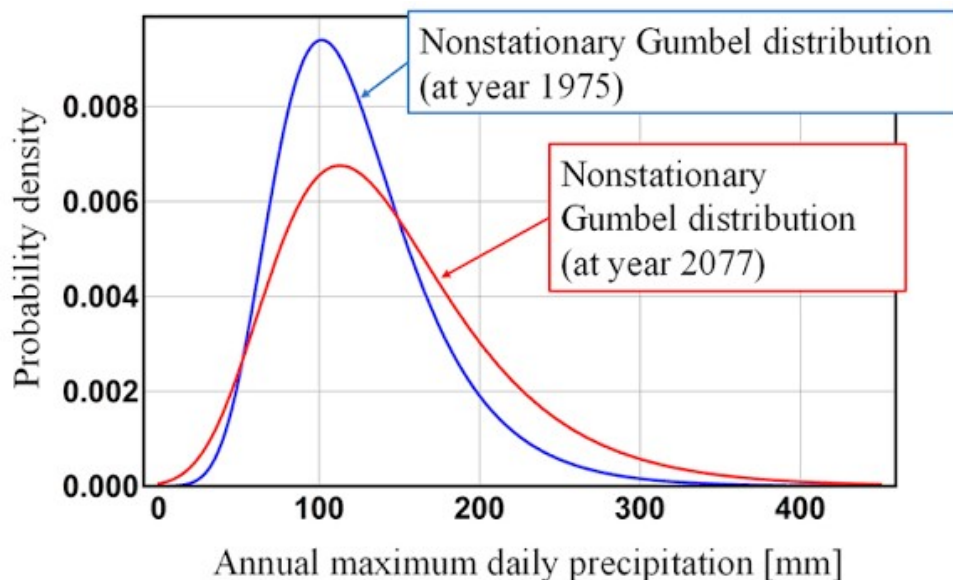
For example, a 100-year flood is a flood having an annual exceedance probability of:

$$P = \frac{1}{100} = 1\%$$

Method of FFA:

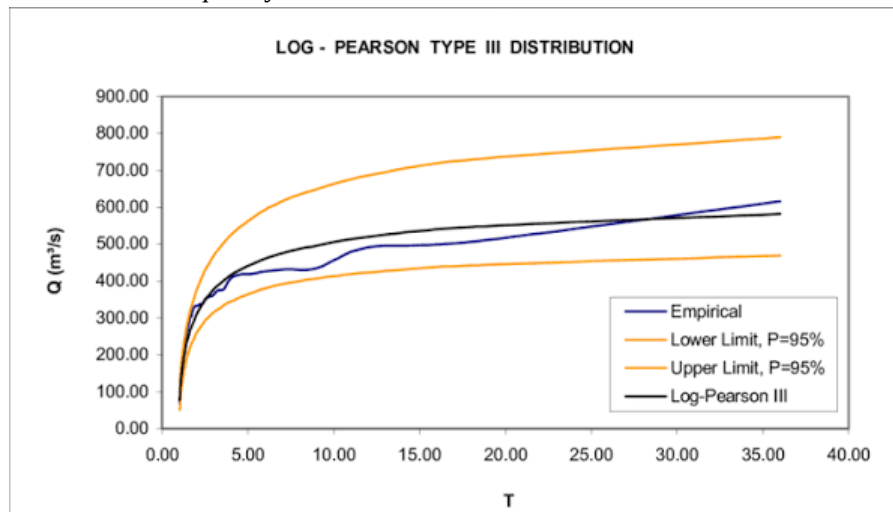
- i. Gumbel's extreme value distribution:

This method of flood frequency analysis Uses a reduced variate derived from the sample mean, standard deviation, and sample size to calculate flood magnitude for a given return period.



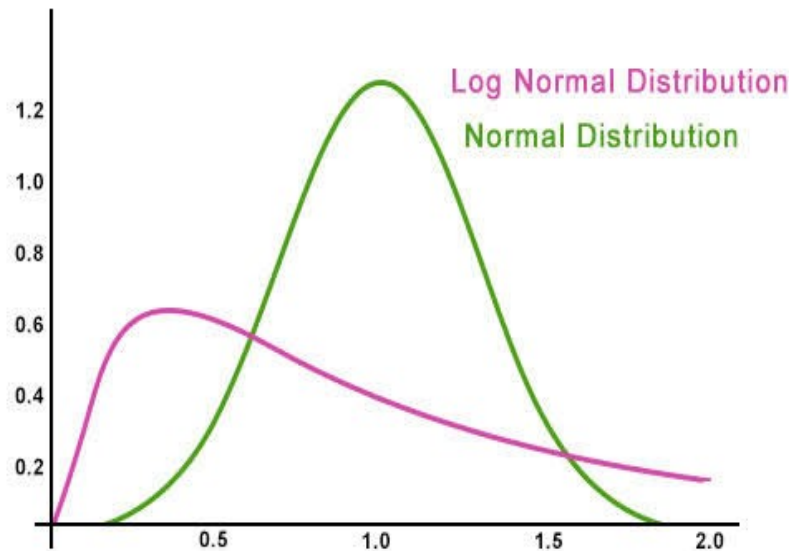
- ii. Log-Pearson Type III distribution:

It is the standard approach that log-transforms the discharge data and incorporates a coefficient of skewness alongside standard frequency factors.



- iii. Log-Normal Distribution:

It assumes that the logarithms of the extreme flow values follow a normal statistical distribution.



Flood frequency analysis is particularly important in hydropower projects as the structures must safely withstand extreme floods. The importance of FFA in hydropower development are listed below:

- i. FFA helps determine the design flood for sizing spillways and ensuring safe passage of floods.
- ii. It is also used to select an appropriate diversion design flood during construction.
- iii. It also provides flood estimates required for checking the safety of dams, weirs and other headworks.
- iv. It also helps determine design flood discharge and waterway requirements for access bridges.
- v. Flood-frequency estimates can be used as input for routing floods through reservoirs and river reaches.
- vi. It helps balance the cost of flood-protection structures against the risk of flood damage.

## 2. Explain the various method of determining flood discharge in a stream.

Answer:

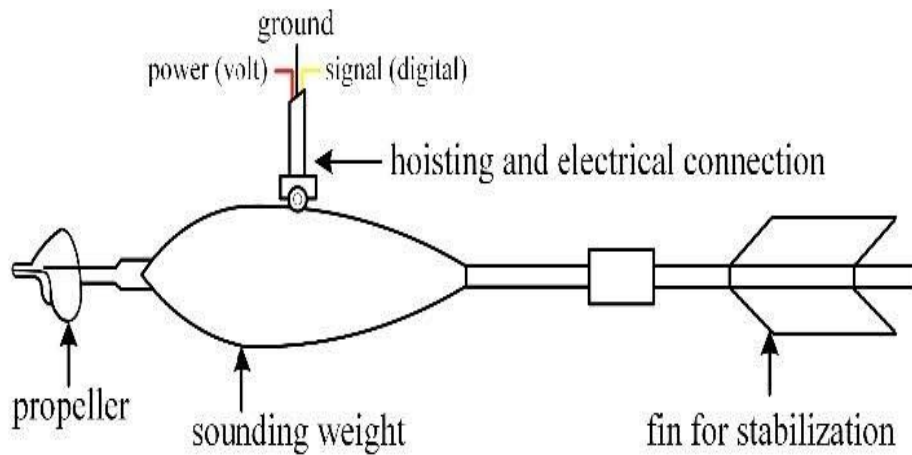
The quantity of water flowing through a stream during a flood, usually expressed in  $\text{m}^3/\text{s}$  is known as flood discharge. Accurate estimation of flood discharge is essential for the design of bridges, culverts, spillways, dams, flood-control structures and other hydraulic works.

The methods for determining discharge in a stream can be broadly classified into direct field measurements and indirect analytical estimations.

### i. Direct Measurement Methods:

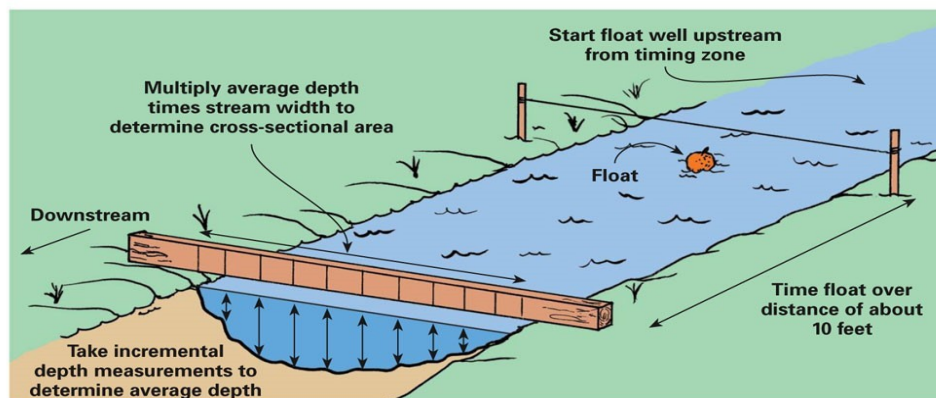
These methods involve measuring physical parameters of the stream flow directly at the site.

- a. **Current Meter (Area-Velocity Method):** This is the standard method for computing discharge. The stream cross-section is divided into several vertical strips, and a current meter is used to measure the velocity ( $V$ ) at each vertical. The total discharge ( $Q$ ) is the sum of the discharges from each strip, calculated as:  $Q = \sum (\text{Area} \times \text{Velocity})$ .



- b. **Float Method:** For simpler or less accurate measurements in small streams, a floating object is timed as it travels a known distance to determine surface velocity.

## The Float Method of Estimating Flow



- c. **Weirs and Flumes:** Impervious barriers such as weirs or shaped channels such as flumes are used for small to medium discharges. These structures raise the upstream water level, and the discharge is calculated based on the head of water flowing over the crest.

### ii. Indirect and Analytical Estimation Methods

These techniques use historical data, catchment characteristics, or physical evidence of past floods.

- a. **Rational Method:** This is suitable for catchments with limited area. The peak discharge is calculated using the formula:  $Q = 360CIA$  Where  $C$  is the runoff coefficient,  $I$  is rainfall intensity (mm/hr), and  $A$  is the catchment area (ha).
- b. **Empirical Formulas:** These are generally used for large catchment areas. Common formulas practiced in Nepal include:
  - a. **Dicken's Formula:**  $Q = CA^{3/4}$ .
  - b. **Modified Dicken's formula:**  $Q = C_t A^{3/4}$
  - c. **Ryve's Formula:**  $Q = CA^{2/3}$
- c. **Slope-Area Method:** This indirect technique computes peak flood discharge using high-water marks, channel geometry, and Manning's roughness equation. It is particularly useful for estimating discharges after a flood has passed by measuring the water surface slope and wetted perimeter.
- d. **Unit Hydrograph (UH) Method:** A unit hydrograph represents the direct runoff resulting from a unit depth (1 cm) of excess rainfall occurring uniformly over the basin. It acts as a unit pulse response function to estimate flood hydrographs for various storm durations and intensities.

### iii. Continuous Monitoring and Forecasting

- a. **Stage-Discharge (Rating Curve) Method:** A **rating curve** is a graphical relationship between the water level (**stage**) and the corresponding discharge. Once established through multiple field measurements, it allows for continuous discharge monitoring by simply reading the gauge height. The relationship is typically expressed as:  $Q=C(H-a)^n$ .
- b. **Probable Maximum Flood (PMF):** This is the extreme flood physically possible in a region, determined by maximizing the worst meteorological and hydrological conditions. It is used for designing critical structures where failure would be catastrophic.

iv. **Computational Techniques**

Modern engineering often employs software for more complex modeling of stream discharge. Frequently used software in Nepal includes HEC-RAS for river modeling, HEC-HMS for watershed analysis, and ArcGIS/QGIS for spatial data assessment.

In hydrology and water-resources engineering, WECS, DHM, and MIP are methods are commonly encountered in Nepal for estimating flood discharge or hydrological design parameters.

i. **WECS Method:**

WECS stands for Water and Energy Commission Secretariat.

WECS method is an empirical regional method developed for estimating flood discharge in ungauged catchments of Nepal.

It uses catchment characteristics such as catchment area and regional hydrological relationships to estimate design floods.

ii. **DHM Method:**

DHM stands for Department of Hydrology and Meteorology, Nepal.

The DHM method refers to flood estimation procedures/empirical relationships developed or used by DHM based on observed rainfall and river-flow data from Nepalese gauging stations.

DHM provides hydrological data such as rainfall, river discharge, water levels and flood information, which are used in design.

iii. **MIP Method:**

MIP generally refers to the Modified Index Flood (MIP) method in regional flood estimation.

The method is based on the index-flood concept, where flood characteristics from gauged catchments are transferred to an ungauged or poorly gauged catchment using regional relationships.

3. **a. What are the forms of energy contained in flow through pipes? Explain with the help of Bernoulli's equation in a neat sketch.**

**b. How are losses in head in pipe contractions, expansions and bends expressed and estimated?**

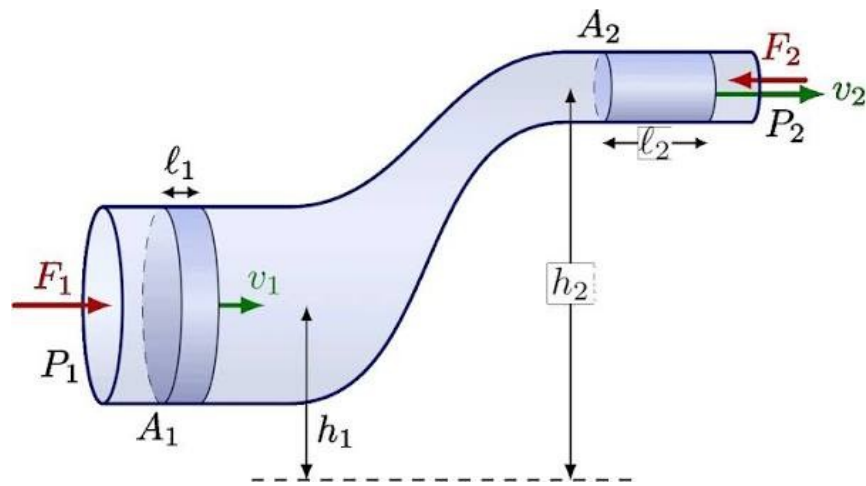
Answer:

a. For a flowing fluid in a pipe, the total energy at any section consists mainly of three forms of energy:

- i. Pressure energy
- ii. Kinetic energy
- iii. Potential energy

According to Bernoulli's theorem, for steady, incompressible and frictionless flow, the total energy remains constant along a streamline.

$$\frac{P}{\gamma} + \frac{V^2}{2g} + Z = \text{constant}$$



i. Pressure head:

- The energy possessed by water due to its pressure is called pressure energy, also known as pressure head.
- It is expressed as:  $\frac{P}{\gamma}$

ii. Kinetic energy:

- The energy possessed by water due to its velocity is called kinetic energy, also known velocity head.
- Kinetic energy per unit weight is:  $\frac{V^2}{2g}$

iii. Potential energy:

- The energy possessed by water due to its elevation above a chosen datum is called potential energy.
- Potential energy per unit weight is:  $z$

b. When water flows through pipes, changes in pipe size or direction cause local or minor head losses due to turbulence, separation and changes in velocity. In general, local head loss is expressed as:

$$h_L = K \frac{V^2}{2g}$$

where,

$h_L$  = head loss,

$K$  = loss coefficient,

$V$  = appropriate mean velocity, and

$g$  = acceleration due to gravity.

i. Loss due to sudden expansion:

When a pipe suddenly increases in diameter, flow separates from the pipe wall and eddies are formed. The head loss is given by:

$$h_{Le} = \frac{(V_1 - V_2)^2}{2g}$$

ii. Loss due to sudden contraction:

When a pipe diameter suddenly decreases, the flow contracts to form a vena contracta, followed by turbulence and energy loss.

$$h_{Lc} = K_c \frac{Vc^2}{2g}$$

The value of  $K_c$  depends mainly on the ratio of pipe areas/diameters and the geometry of the contraction.

iii. Loss due to bends:

When water flows through a bend, a change in direction causes turbulence, secondary currents and separation, producing head loss.

$$h_{Lb} = K_b \frac{V^2}{2g}$$

where  $K_b$  is the bend-loss coefficient.

4. **In Nepalese context, draw your opening on comparative benefit of each Thermal, Hydro, Wind and solar power.**

Answer:

Nepal is one of the richest countries in term of natural resources. It has good potential for hydropower, solar and wind energy, while thermal power has a more limited role because of its dependence on imported fuel.

i. Hydropower:

- a. It is the electricity generated by utilizing the energy of flowing rivers.
- b. Globally, hydropower covers about 14.3% of total electricity generation.
- c. Nepal has more than 6000 big and small rivers which has net electricity generation capacity of 83000 MW making it a second richest country in water resources after Brazil.
- d. Due to steep topography that provides the required head for power generation and availability of fast flowing rivers in Nepal, hydropower is most suitable option of alternate energy system.

ii. Solar power:

- a. It is the electricity generated by utilizing the sunlight.
- b. Powered by the photovoltaic effect (using silicon solar cells) or solar-thermal technology, it provides a renewable, clean, and increasingly cost-effective way to power homes, businesses, and entire utility grids.
- c. Solar energy covers about 10% of global electricity demand.
- d. It is ideal for remote mountain regions with high solar radiation where other alternate source of energy is difficult to supply or are expensive.

iii. Wind power:

- a. Wind energy is a renewable power source that utilizes the kinetic energy of moving air to generate electricity.
- b. Potential of wind energy exists in the district of Nepal with the presence of high velocity wind such as Mustang, Palpa, Dolpa, and other high-wind regions.
- c. It is currently limited in application.
- d. Can complement hydropower, particularly where wind availability differs seasonally from river flow.
- e. Suitable for small and decentralized power systems where site conditions are favorable.

iv. Thermal power:

- a. Can provide relatively reliable and controllable generation independent of rainfall.
- b. Useful as backup during hydropower shortages and emergencies.
- c. Can be located relatively close to major load centers.
- d. Construction can be faster than large hydropower projects.



- e. However, Nepal's dependence on imported fossil fuels makes it comparatively expensive and less environmentally desirable.

5. **Explain the procedure for obtaining a “License for electricity generation” of a 100 MW run of river hydropower project as per the provisions of the Electricity Act, 2049 and Electricity regulation, 2050.**

Answer:

To obtain a License for Electricity Generation for a 100 MW run-of-river hydropower project, an applicant must follow the procedures established in the Electricity Regulation, 2050 (1993), as detailed below:

i. **Submission of Application:**

Any person or corporate body intending to produce electricity must submit an application in triplicate to the Secretary of the Ministry of Energy through the Department of Electricity Development. The application must follow the format prescribed in Schedule-6 and include the following particulars:

- **Project Details:** A detailed description of the project, including a site map, the source of electricity, estimated cost, construction timeline, and details of partners or directors.
- **Feasibility Analysis:** Technical and economic analysis, description of potential consumers, and estimated quantity of electricity to be sold.
- **Mode of Finance:** Detailed financial plan, including the economic condition of investors, commitments from financial institutions, and the debt-to-equity ratio.
- **Land and Environmental Impact:** Description of public or private land to be acquired and an **environmental effect analysis**. This analysis must detail measures to minimize adverse effects, local benefits and plans for the rehabilitation of evacuated people.
- **Power Sale and Transmission:** Description of the proposed transmission line and any agreements or letters of intention regarding the sale and purchase of electric power.

ii. **Examination of Application:**

Once received, the Secretary or the Department examines the application to ensure all necessary documents and reports are included. If any documents are missing, the applicant is notified within 45 days and given a reasonable time limit to submit the required information.

iii. **Public Notice and Reaction:**

After the initial examination, the Department must publish a public notice containing the project's details for the information of the general public.

- **Public Reaction:** Any person may provide feedback or state reasons why the project might cause adverse effects within 35 days of the notice's publication.
- **Condition Setting:** If any reactions are received, the Secretary will consider them and may include specific conditions in the license to address those concerns.

iv. **Issuance of the License:**

If the application is deemed satisfactory after the examination and public notice period, the Secretary issues the license in the format prescribed in Schedule-9(A).

- **Existing Survey Licensees:** If the applicant already holds a survey license and applies within its validity period based on their own survey, the production license shall be issued subject to regular provisions.
- **Legal Compliance:** The applicant must also ensure they are registered under the Industrial Enterprises Act, 2049 and, if foreign investment is involved, have obtained permission under the Foreign Investment and Technology Transfer Act, 2049.

v. **Fees and Commencement of Work:**

- **License Fee:** For a project with a capacity of 100 MW, the license fee for the initial period is Rs. 500,000.
- **Commencement:** The licensee must start physical construction works within one year of obtaining the license and inform the Department of Electricity Development.
- **Reporting:** Once works begin, a progress report must be submitted to the Department every six months until completion.

6. **a. What is meant by:**

- i. **Load factor**
- ii. **Secondary energy**

### iii. Firm energy      iv. Seasonal energy

#### b. How can secondary energy be utilized?

##### i. Load factor:

- It is the ratio of average load to maximum load.
- It gives an idea of the cost of electricity per unit power generation.
- Higher the value of load factor, denotes less maximum demand. Less maximum demand can be created by a low-capacity power plant.
- As the capacity of plant is low, the initial as well as running cost will also be low. Thus, the cost per unit power generation will be less.
- Higher value of load factor means higher average load. this means the plant is operating near its rated capacity. Therefore, the cost per unit generation will be less.

##### ii. Secondary energy:

Secondary energy is the electrical energy generated by a hydropower plant in excess of its firm energy, usually during periods when sufficient water is available but the energy is not guaranteed throughout the year.

In a run-of-river hydropower plant:

- During the dry season, river flow is low such that the plant produces mainly firm energy.
- During the monsoon, river flow increases and the electricity generation can exceed the firm-energy output. This additional generation is called secondary energy.

Total energy = Firm energy + Secondary energy

##### iii. Firm energy:

Firm energy is the amount of electrical energy that a hydropower plant can reliably and continuously generate during the critical low-flow period, when the available water is at its minimum. It is also called dependable or guaranteed energy.

- In a run-of-river project, firm energy is mainly determined by dry-season river flow.
- In a storage project, reservoir storage can be used to maintain firm generation during low-flow periods.

Firm energy is generally less than the total annual energy generated by the plant.

##### iv. Seasonal energy:

Seasonal energy is the energy generated by a hydropower plant during a particular season when water availability is high, generally during the monsoon or wet season.

It varies throughout the year because it depends on seasonal variation in river flow.

In Nepal:

- During dry season, the river discharge becomes low where mainly firm energy can be generated.
- During monsoon season, the river discharge increases significantly where hydropower plant can generate additional seasonal energy.

b. Secondary energy is the electrical energy generated by a hydropower plant in excess of its firm energy, usually during periods when sufficient water is available but the energy is not guaranteed throughout the year.

It can be utilized in following ways:

- i. Surplus power can be supplied to the grid to meet the additional demand during periods of high generation.
- ii. It can be used for industries, factories and other energy-intensive activities, especially during periods of surplus generation.
- iii. It can be used to operate irrigation pumps and water-lifting systems, during wet seasons.
- iv. It can be used for electric heating, boilers and other thermal applications.
- v. It can support electric railways, electric vehicles and other electric transportation systems.

#### 7. **Classify the embankment dams. Discuss the stability of an embankment dam.**

Answer:

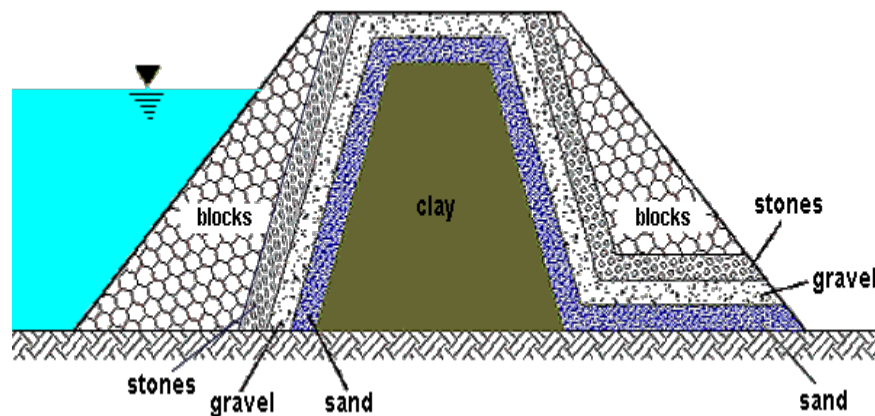


An embankment dam is a dam made up of compacted earth, rock, or a combination of these materials. It resists water pressure primarily through its mass, shear strength and geometry.

Embankment dams can be classified into:

i. Earth-fill dams:

These dams are constructed by using compacted soil. they are further divided into:



a. Homogeneous earth dam:

It is constructed entirely or almost one type of the earth materials suitable for low height dams only.

b. Zoned earth dam:

It is also known as non-homogeneous dams. These dams are constructed with the materials of different kinds in the different part of the embankment.

c. Diaphragm earth dam:

This type of dams is constructed with pervious material with thin impervious diaphragm in the central part to prevent leakage of water.

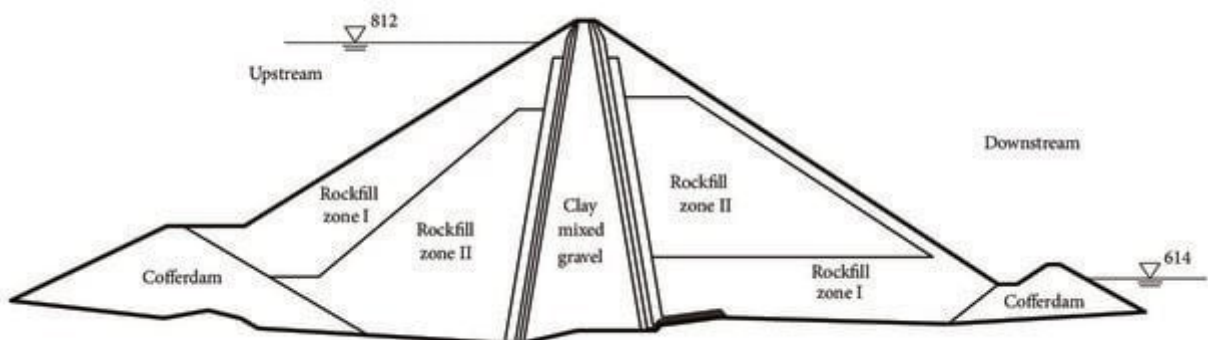
Example: embankment dam project at Rupa lake.

ii. Rock-filled dams:

- An embankment is said to be rock filled dam if fill material is coarse grained gravel, crushed rock or boulders with more than 50% of the total placed volume of material.
- In this type of dam, there is an impervious core of compacted earth fill or a slender concrete or bituminous membrane.
- In Nepal, some of the most popular rock-filled dams are Kulekhani dam, Dhaph dam and so on.

iii. Composite dams:

These types of composite dams use both earth and rock materials in different zones according to their engineering properties.



The stability of an embankment dam is governed by its ability to resist sliding, seepage, erosion and deformation under different loading conditions.

i. Stability against slope failure:

The factor of safety against sliding is expressed as:

$$FoS = \frac{\text{Resisting Forces (Shear Strength)}}{\text{Driving Forces (Shear Stress)}}$$

Where:

*Resisting forces = Cohesion  $\wedge$  friction of soil/rock.*

*Driving forces = Self-weight of embankment*

*water pressure*

*seismic forces acting downslope.*

If  $FoS > 1.5$ , then the dam is said to be stable.

If  $FoS \approx 1.0$ , then it is said to be in Critical  $\vee$  limiting equilibrium

If  $FoS < 1.0$ , then the dam becomes unstable.

ii. Seepage stability:

- Excessive seepage in embankment dams can cause piping, erosion, loss of water, reduction in effective stress and slope stability etc.
- Hence, seepage must be controlled using impervious cores, cutoff, filters, drainage blankets, toe drains etc.

iii. Stability against piping:

- Seepage force should not be sufficient to carry soil particles through the dam or its foundation.
- Properly designed filters and transition zones prevent migration of fine particles while allowing seepage water to drain safely.

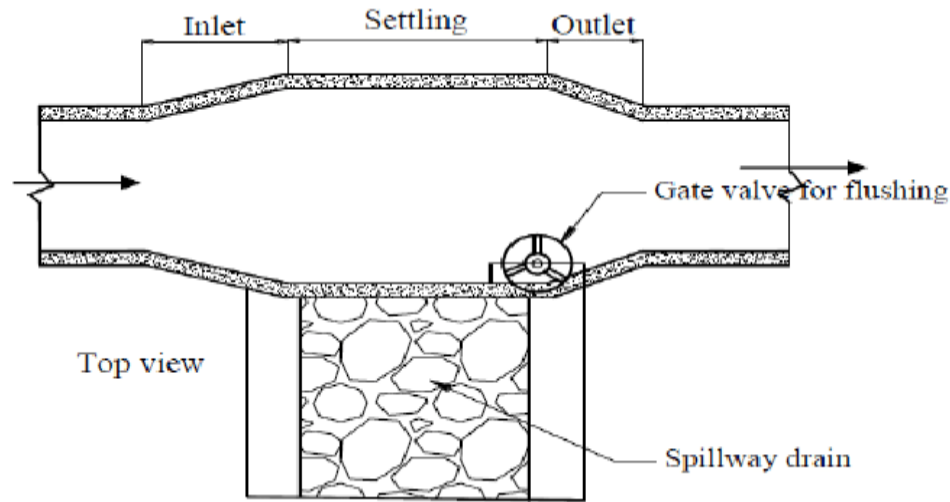
iv. Stability against surface erosion:

- Upstream slope of embankment dams may be damaged by wave action, while the downstream slope may be affected by rainfall and surface runoff.
- Protection is provided by: riprap or stone pitching, turfing, proper surface drainage etc.

**8. Explain the major design criteria of De-sanding basin. For conventional settling basin, explain why the inlet angle of basin is much smaller than that of outlet? Explain with neat sketch.**

Answer:

De-sanding basin also known as de-stilling basin is a hydraulic structure placed between intake and penstock in the hydropower project for removal of suspended sediment particles from flowing water which cannot be trapped by the gravel trap.



The major design criteria of de-sanding basin are listed below:

**i. Design discharge:**

- De-sanding basin must be designed for the maximum expected flow rate.
- The capacity of de-sanding basin should be adequate to handle peak flows without reducing its efficiency.

**ii. Size of particles:**

- The size of particles required to be removed must be selected.
- Generally, particles with its size greater than 0.15-0.2 mm are targeted for removal.

**iii. Settling velocity of particles:**

- It can be determined using stoke's law.

$$V_s = \frac{g(\rho_s - \rho)d^2}{18\mu}$$

Where:

$V_s$  = settling velocity

$d$  = particle diameter

$\rho_s$  = density of particle

$\rho$  = density of water

$\mu$  = dynamic viscosity

**iv. Basin dimensions:**

- The length should be adequate for complete settling of particles.
- Length-to-width ratio is generally kept between 4:1 and 8:1.

In a conventional de-sanding basin, the inlet transition is made gradual, whereas the outlet transition can be comparatively steeper.

At the inlet, water enters into de-sanding basin with relatively high velocity. A small inlet angle provides a gradual expansion of the flow section which reduces sudden turbulence, eddies and separation.

At the outlet, the water passed through the settling zone and has relatively low velocity. A larger outlet angle helps to collect the clarifies water and direct it toward outlet.

Hence, inlet angle is provided smaller than outlet angle because the inlet requires more gradual transition to minimize the turbulence and ensure uniform flow distribution.

**9. Explain types and advantages of underground power stations.**

Answer:

An underground power station is a hydropower station in which the powerhouse and major equipment are constructed below ground level.

Underground power stations can mainly be classified based on the arrangement of the powerhouse and waterways:

- i. **Completely underground power station**
  - The powerhouse, turbines, generators and associated equipment are located entirely underground.
  - It is suitable where surface space is limited or geological conditions are favorable.
- ii. **Partly underground power station**
  - Some components, generally the powerhouse or auxiliary structures, are underground while other facilities remain on the surface.
  - It provides a compromise between underground and conventional surface arrangements.
- iii. **Underground cavern power station**
  - The powerhouse is constructed inside a large underground cavern excavated in competent rock.
  - It is commonly used for large hydropower projects and may contain several generating units.

**Advantages of underground power stations:**

- i. They are particularly suitable for steep and mountainous regions, such as Nepal, where suitable flat land for a surface powerhouse may be unavailable.
- ii. Very little surface land is required, reducing land acquisition and disturbance.
- iii. Underground structures are relatively well protected from landslides, avalanches, floods, storms and other surface hazards.
- iv. Equipment is protected from extreme weather conditions and temperature variations.
- v. Surface disturbance, visual impact and destruction of vegetation can be minimized.
- vi. Being underground provides additional protection against external damage and unauthorized access.
- vii. Underground arrangements are particularly advantageous where a long headrace tunnel and steep hydraulic gradient are involved.

**10. How can we classify turbines? Indicate their suitability for different range of heads and specific speed, also discuss other important factors that are considered in the selection of turbines.**

Answer:

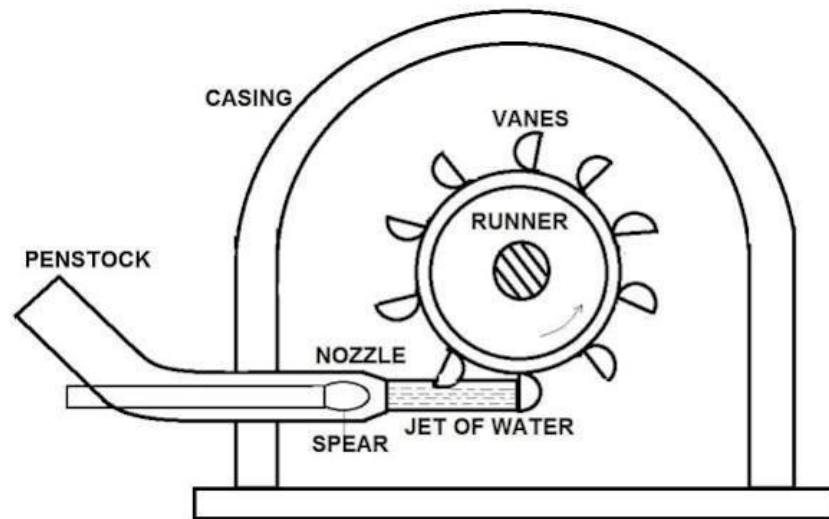
Turbine is the hydro-mechanical equipment that converts hydraulic energy into mechanical energy. It consists of wheel called runner provided with the number of curved vanes on the periphery. And guiding apparatus to direct the flow of water at the inlet in specified direction. The selection of a hydraulic turbine depends on the available head, discharge, power output, efficiency, operating conditions, and economic considerations. Selection of turbine is done such that it gives maximum efficiency, reliable operation, and minimum maintenance cost.

We can classify the turbines as follows:

**i. On the basis of type of energy at inlet:**

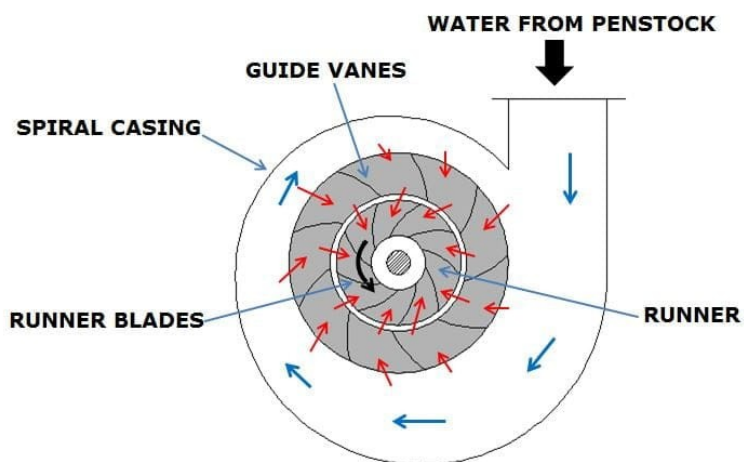
**a. Impulse turbine:**

- The conversion of energy is due to change in momentum of water or due to the change in kinetic energy only is called impulse turbine.
- Water enters and leaves the turbine at atmospheric pressure in impulse turbine.
- It is suitable for high head and low discharges.
- Buckets are in action only when they are in front of the nozzle.
- Example: Pelton turbine



b. Reaction turbine:

- The conversion of energy is due to the pressure difference at the two sides of vanes or due to the change in pressure of water.
- Water enters the turbine at high pressure and leaves the turbine at low pressure.
- It is suitable for low to medium head and medium to high discharges.
- The vanes are in action all of the time in reaction turbine.
- Example: Francis turbine, Kaplan turbine etc.



ii. On the basis of flow directions:

a. Tangential flow turbine:

- The direction of flow is tangential to the shaft in tangential flow turbine.
- Example: Pelton turbine

b. Axial flow turbine:

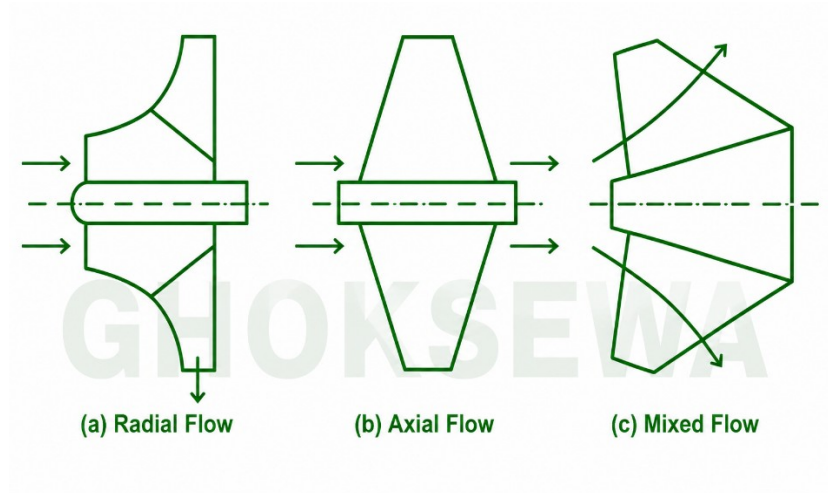
- The direction of flow is parallel to the shaft in axial flow turbine.
- Example: Kaplan turbine

c. Radial flow turbine:

- The direction of flow is radial in radial flow turbine.
- Example:

d. Mixed flow turbine:

- The entry of flow is radial and exit is axial in mixed flow turbine.
- Example: Francis turbine



### iii. On the basis of specific speed:

#### a. Low specific speed turbine:

- The turbine whose specific speed is less than 50 rpm is known as low specific speed turbine.
- Example: Pelton turbine



#### b. Medium specific speed turbine:

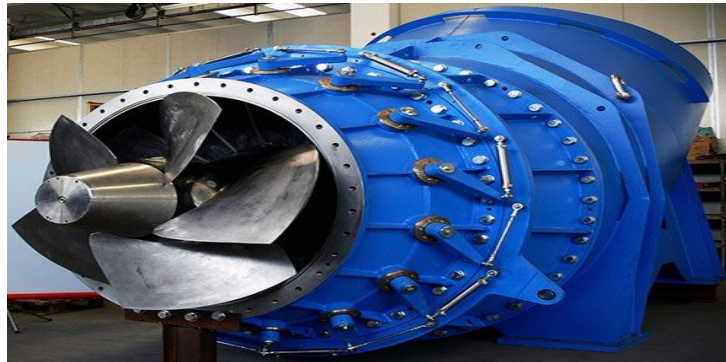
- The turbine whose specific speed is equal to 50 to 260 rpm, is known as medium specific speed turbine.
- Example: Francis turbine





c. High specific speed turbine:

- The turbine whose specific speed is greater than 260 rpm, then it is known as high specific speed turbine.
- Example: Kaplan turbine



The suitability for different range of heads and specific speed are listed below:

- i. Pelton turbine:
  - The turbine is suitable for high head, generally greater than 250 m.
  - It has low specific speed, making its application suitable in mountainous areas, high-head hydropower.
- ii. Francis turbine:
  - It is used for medium head about 30-250 m.
  - It has medium specific speed, suitable in medium-head hydropower plants.
- iii. Kaplan turbine:
  - Kaplan turbine is suitable for low head, generally less than 30 m.
  - It has high specific speed, making it suitable for large discharge, low head river sources.
- iv. Propeller turbine:
  - Propeller turbine is used for very low head hydropower project, generally less than 30 m.
  - The specific speed is very high, suitable for low-head or constant-load conditions.

The other important factors that are considered in selection of turbines:

- i. If the available head varies considerably, a turbine is capable of maintaining good efficiency over the range that should be selected.
- ii. The turbine should have high efficiency under normal operating conditions and preferably maintain the good efficiency during partial-load conditions.

- iii. The possibility of cavitation should also be considered. Proper turbine type must be selected to minimize cavitation.
- iv. The turbine speed should be compatible with the required generator speed.
- v. Both initial and operating cost should be considered, as the selected turbine must be economical.

In Nepal, **Francis turbine** is by far the most commonly used turbine type in Nepal's hydropower projects because most of the rivers in Nepal have medium to high heads (roughly 60–350 m) with moderate to high discharge, the conditions where Francis turbines perform best.